Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High thermal conductive passive cooling components of composite Material with complex 3D shape			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (x), Secondary Battery ( ), Advanced Mobility (x), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (x), Hydrogen ( ), Cybersecurity ( ), AI (x), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy and Resources		Energy and Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Transmission & Distribution Systems	Power Electronics Technologies	Transmission & Distribution Systems	New Materials for Power Systems
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The ongoing miniaturization and increasing power density in microelectronic and power electronic systems impose severe challenges on thermal management and packaging technologies. Conventional cooling solutions based on aluminum or copper reach their physical and geometrical limits, particularly for highly integrated and application-specific designs. Copper–diamond composites represent a highly promising material class, combining the excellent thermal conductivity of diamond with the processability of copper. Thermal conductivities of up to 700 W/(m·K) at room temperature have been reported for planar copper–diamond composites with high diamond volume fractions. However, their application has so far been limited to simple geometries due to processing constraints.			

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	In this project, copper–diamond composites with approximately 60 vol.% diamond are transferred into the additive manufacturing domain to enable three-dimensional, application-specific cooling components. A novel processing route combining gel casting for shaping and a quasi-isostatic Spark Plasma Sintering (SPS) approach for densification is applied. After fabrication of the green body, debinding and pre-sintering under hydrogen atmosphere are performed, followed by quasi-isostatic SPS to achieve high density and tailored thermal properties. This approach enables the realization of complex, near-net-shape thermal management components with outstanding thermal conductivity and reduced density. The resulting components open up new design freedoms for direct and efficient cooling in demanding applications such as microelectronics, power modules, charging infrastructure, e-mobility systems, aerospace electronics and AI hardware. Furthermore, the developed processing strategy is transferable to other high-thermal-conductivity composite systems, increasing its industrial relevance.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* The successful development and industrial implementation of complex-shaped copper–diamond cooling components requires a close and iterative collaboration between a research institute and an industrial partner. While the research institute contributes advanced materials expertise and novel processing technologies, the industrial partner provides application-driven requirements, boundary conditions and validation scenarios. Thermal management solutions are highly application-specific with respect to geometry, interface conditions, operating temperatures and reliability requirements. Therefore, the industrial partner defines the target use cases, component specifications and performance criteria. Based on this input, the research partner adapts and optimizes the processing route, material composition and component design. The joint R&D project follows a two-step evaluation strategy. In the first step, component geometry, microstructure and thermal performance are characterized and validated at the research institute using standardized test setups and simulated heat sources. In the second step, the components are evaluated under realistic application conditions by the industrial partner. This iterative loop enables rapid optimization of the material–process–design combination and significantly reduces development risk for later industrial adoption.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Copper–diamond composites 2. High-thermal-conductivity materials 3. Additive manufacturing of complex geometries 4. Thermal management for microelectronics and power electronics 5. Next-generation cooling components / cold plates

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* The overall objective of this joint R&D project is the development of **additively manufactured, three-dimensional copper–diamond cooling components** with outstanding thermal performance and application-specific geometries. Specific objectives include: • Development of a robust and scalable shaping process for complex copper–diamond green bodies. • Optimization of quasi-isostatic Spark Plasma Sintering parameters to achieve high density and tailored thermal conductivity. • Design and fabrication of application-oriented prototype components based on industrial requirements. • Experimental validation of thermal performance under laboratory and application-relevant conditions. • Demonstration of the technology's suitability for industrial implementation and further scale-up.			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* The target specifications of the developed copper–diamond components are: • Thermal conductivity: **≥ 650–680 W/(m·K) at room temperature**, depending on geometry • Density: **≤ 5.5 g/cm³** • Relative density after sintering: **≥ 99 %** • Geometrical complexity: Three-dimensional cooling structures e.g. with integrated channels or fins • Component size (demonstrators): application-specific, up to several tens of cm³ • Thermal stability: Stable performance under operating temperatures relevant for power and microelectronics applications			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* The R&D project is structured into two phases: **Phase 1 – Pre-project (6 months)** • Definition of target applications and component requirements together with the Korean SME • Conceptual design of demonstrator components • Adaptation of copper–diamond feedstock and gel casting formulations • Initial fabrication of simple 3D demonstrators • Preliminary thermal and microstructural characterization **(MS1: defined properties are met)** **Phase 2 – Main project (28 months)** • Optimization of shaping and debinding processes for complex geometries • Development and optimization of quasi-isostatic SPS densification routes • Fabrication of application-specific prototype components • Detailed characterization of microstructure, density and thermal conductivity **(MS2: 3D complex shaped demonstrator with less than 10 % deviation from MS1 properties)** • Laboratory-scale thermal performance testing using simulated heat sources • Validation under application-relevant conditions at the industrial partner **(MS3: increased performance as defined)** • Iterative optimization based on test results • Assessment of scalability and industrial implementation potential			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* The research institute will be responsible for: • Development and optimization of copper–diamond composite formulations • Design and implementation of the gel casting shaping process • Development and optimization of quasi-isostatic SPS processing routes • Microstructural, physical and thermal characterization of materials and components • Laboratory-scale thermal performance evaluation • Scientific documentation and dissemination of results
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* The Korean SME partner will be responsible for: • Definition of application scenarios and component requirements • Contribution to component design from an industrial perspective • Provision of boundary conditions and operating requirements • Validation of prototype components under application-relevant conditions • Feedback for iterative optimization of materials and component design • Evaluation of commercialization potential and industrial scale-up

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* The expected outcomes of the joint R&D project include: • A novel manufacturing route for complex-shaped copper–diamond cooling components • Demonstration of significantly enhanced thermal management performance compared to conventional solutions • Application-specific prototype components validated under realistic conditions • Reduced development risk and time-to-market for industrial partners From an industrial and economic perspective, the project enables Korean SMEs to access cutting-edge thermal management technology with strong differentiation potential in high-growth markets such as power electronics, AI hardware and e-mobility. At the same time, the project strengthens international collaboration and establishes a basis for long-term technology transfer and commercialization.

6. References

(1) Schubert, Thomas, T. Weißgärber, and Bernd Kieback. "Fabrication and properties of Copper/Carbon composites for thermal management applications." Advanced Materials Research 59 (2009): 169-172.

(2) Hutsch, Thomas, et al. "Copper-Diamond Composite With Complex Shape By Gel Casting Plus Spark Plasma Sintering." *Euro Powder Metallurgy Congress and Exhibition 2024*. 2024.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization